

scDesignGuard: source-bound design gating for multisample single-cell differential analysis

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Abstract

Single-cell workflows can produce numerical results even when source identity, biological replication, target support or model estimability are unresolved. We developed scDesignGuard, a source-bound stage-aware framework, and implemented a deterministic evaluator that assigns PROCEED, ABSTAIN, BLOCK or NON_EVALUABLE to a preconstructed design contract before downstream inference. Against a frozen source-bound human consensus in an overlap-excluded benchmark of 15 study families, scDesignGuard cleared 4/4 valid designs, blocked 6/6 hard-invalid designs and contained 5/5 unresolved designs, with no unsafe continuation. In four overlap-excluded end-to-end cases, two reached model validity and two stopped at explicit source-binding or donor-replication gates. Nine controlled challenge perturbations produced the prespecified state and reason. The checksum-bound evaluator reproduced in source and wheel fixture checks. Applied to 99,029 pancreatic beta cells from 48 donors, the framework localized same-cohort type 2 diabetes-associated composition, transcriptional and competitive gene-set-enrichment signals.

Single-cell workflows can complete and return apparently precise results even when the experimental unit or requested contrast is scientifically unsupported. Computational reviews have documented both the power and limitations of scRNA-seq analysis pipelines.¹ Data integration and atlas construction introduce harmonization and reference-selection challenges.^{2–4} Healthy-reference strategies address the distinct problem of identifying disease-altered states against an appropriate baseline.⁵ Extensible ecosystems make cross-study workflows easier to run.⁶ Yet technical executability does not establish inferential validity. Best-practice guidance treats donors or samples—not cells—as the biological replicates in multisample studies.⁷ When cells are analysed as independent replicates, pseudoreplication can inflate significance and false discoveries.^{8–10}

Moving inference to the biological unit is therefore central to multisample analysis. Donor-aware pseudobulk methods generally provide better-calibrated tests than cell-level analyses.^{8–9,11} Cross-study benchmarks show that integration strategy can alter differential-expression results.¹² Individual-level and multisubject differential-expression models explicitly represent between-subject replication.^{13–14} For abundance endpoints, differential-composition methods use complementary frequentist, Bayesian and neighbourhood-based formulations.^{15–17} Flexible-design and variability-aware models extend composition analysis beyond simple two-group settings.^{18–19} For the design problem addressed here, these approaches begin after a scientifically coherent task has been defined; scDesignGuard separately asks whether the source object, donor map, target semantics and estimand jointly authorize that task.

Methodological work also shows that transformation choices can affect single-cell comparisons.²⁰ Latent multivariate regression extends analysis across multiple conditions.²¹ Neighbourhood-based differential expression, alternative testing paradigms and differential-detection workflows further broaden multisample comparisons.^{22–24} In parallel, simulation studies show that benchmark conclusions depend on whether data-generating mechanisms preserve

realistic count structure and gene correlations.^{25–27} Dataset imbalance can also change apparent integration performance.²⁸ Benchmark platforms organize comparable analytic tasks and evaluation workflows.²⁹ For the design problem addressed here, their common starting point is a declared analysis; scDesignGuard evaluates whether the available evidence supports that declaration.

Reproducible infrastructure solves another part of the problem. FAIR software principles define how computational artefacts should be made findable and reusable.³⁰ Research-object packaging and run-provenance records provide machine-readable evidence chains.^{31–32} Workflow guidance formalizes reproducible computational practice.³³ Workflow registries support discovery and reuse.³⁴ Portable workflow systems support repeatable execution.^{35–36} CELLxGENE and GEO provide persistent dataset identities and repository metadata.^{37–38} BioStudies/ArrayExpress and the Human Cell Atlas broaden repository and atlas coverage.^{39–40} Cell Ontology, OBO Foundry and SSSOM support interoperable target semantics and explicit mapping relations.^{41–43} In the stage-aware framework developed here, these resources supply provenance, portability and semantic evidence, whereas the evaluator separately distinguishes demonstrated invalidity from missing evidence and determines stage-specific authorization.

Here we present scDesignGuard, a source-bound, stage-aware framework, together with a deterministic evaluator for completed design contracts. The evaluator maps reviewed evidence fields to PROCEED, ABSTAIN, BLOCK or NON_EVALUABLE before downstream inference. Semantic fields are constructed upstream from source records and task-specific review. We evaluate the framework in an overlap-excluded benchmark of 15 study families, a separate four-case end-to-end gate evaluation, nine controlled challenge perturbations, multi-repository transport audits and software reproduction. We then apply the gated workflow to a 99,029-cell pancreatic beta-cell atlas.

Results

scDesignGuard converts source-bound evidence into stage-specific authorization

scDesignGuard represents an analysis request as a versioned contract that binds the source object, independent biological unit, condition and pairing structure, target ontology, support rule and estimand (Fig. 1a). These fields are constructed upstream from source records and analysis-specific review. The released evaluator validates the completed contract and maps its supplied evidence fields to PROCEED, ABSTAIN, BLOCK or NON_EVALUABLE with ordered reasons.

In the framework, authorization is sequential (Fig. 1b). Source-scope clearance permits structural-design review; structural clearance permits target-support computation; support clearance permits count aggregation and model-validity assessment. These upstream computations populate the contract and evidence ledger. The released evaluator operates on the resulting contract snapshot, and each passed gate records authorization to assess the next stage. Donor-aware scientific inference begins after final authorization.

GSE144735 illustrates the complete path (Fig. 1c). Its paired contract contained six complete pairs, 3,356 target epithelial cells and 12 donor-condition pseudobulks. The model matrix had rank 7/7 and five residual degrees of freedom. Independent implementations reconciled donor assignment, target-cell counts, library sums, pseudobulk dimensions and model rank before final authorization. The gate evaluation concluded at authorization, before effect estimation. Development support, robustness and calibration evidence is shown in Extended Data Fig. 1.

An overlap-excluded benchmark demonstrates concordant state assignment

The final overlap-excluded benchmark comprised 15 study families with frozen source-bound human consensus labels: four valid, six hard-invalid and five structurally unresolved. scDesignGuard returned PROCEED for all four valid designs, BLOCK for all six hard-invalid designs and NON_EVALUABLE for all five unresolved designs. Valid clearance was 4/4 (95% Wilson CI, 0.510–1.000), hard-invalid detection 6/6 (0.610–1.000), unresolved containment 5/5 (0.566–1.000), unsafe continuation 0/6 (0.000–0.390) and exact state concordance 15/15 (0.796–1.000; Fig. 2).

The source-bound reference review was blinded to evaluator, executable-checklist, end-to-end and biological-outcome outputs. Reviewers verified prepared source-bound fields and froze the human consensus before evaluator execution; benchmark endpoints report concordance with that reference.

End-to-end cases show when analyses proceed and when they stop

A separate overlap-excluded end-to-end cohort comprised four cases processed in frozen order through source scope, structural design, target support, count aggregation, model validity and final authorization (Fig. 3a,c).

GSE144735 contained six complete pairs, 3,356 target cells, 12 pseudobulks, a full-rank 7-column design and five residual degrees of freedom. GSE123813 contained 11 complete pairs, 33,106 target cells, 22 pseudobulks, a full-rank

12-column design and ten residual degrees of freedom (Fig. 3b). Both reached final authorization. Human Cell Atlas project 135f7f5c-4a85-4bcf-9f7c-4f035ff1e428 stopped because exact donor-condition-modality and raw-count-layer binding could not be established. GSE81076 stopped because five total donor identifiers cannot satisfy the frozen minimum of five donors per condition.

Python primary computation, an independently compiled C count implementation and an independent R design-matrix implementation reconciled donor assignment, target-cell counts, raw-library sums, pseudobulk dimensions, rank, condition-effect estimability and residual degrees of freedom. The gate evaluation concluded before differential-expression, pathway and effect estimation.

Controlled challenges and repository audits test boundary handling

Nine deterministic perturbations altered one design property at a time: donor deletion, cell-level pseudoreplication, batch-condition confounding, duplicate donor, pair destruction, wrong target, wrong version, mixed strata or low support. All nine produced a safe non-PROCEED state and the prespecified primary reason (9/9 for both endpoints; Fig. 4a). These prespecified perturbations test rule sensitivity under defined contract defects.

Repository audits covered 18 study families across BioStudies/ArrayExpress, Single Cell Portal, NCBI GEO and Human Cell Atlas (Fig. 4b). At Tier C identity and metadata endpoints, all 18 received NON_EVALUABLE outcomes for unresolved ontology, donor-mapping and support evidence. The audits establish consistent handling of incomplete repository metadata. Details are shown in Extended Data Fig. 3.

The public v0.1.1 release retains the evaluated v0.1.0 wheel, OCI image, JSON Schema and reason registry and provides archive-relative tests, four terminal-state fixtures and checksum-bound receipts. The source suite completed 17/17 test methods with zero failures, errors or skips; four fixture subtests matched their golden states and reasons, and the exact wheel passed 4/4 fixture CLI checks in a clean temporary environment (Fig. 4c). These checks establish deterministic behavior for completed contracts. Synthetic uncertainty and archive status are summarized in Extended Data Fig. 2.

Donor-aware gating localizes same-cohort beta-cell signals

The same-cohort biological application used the GSE221156 pancreatic islet atlas⁴⁴ and comprised 99,029 author-annotated beta cells from 48 donors: 17 normal, 14 prediabetes and 17 type 2 diabetes donors across author-defined clusters C0–C7. Counts were aggregated once per donor–cluster. C7 had the lowest qualifying support (13 normal, eight prediabetes and ten type 2 diabetes donors).

Recovered beta-cell-state composition differed between type 2 diabetes and normal donors after adjustment for assay, sex, ethnicity, age and body mass index (Fig. 5a,b). Mean donor C6 proportions were 0.060, 0.086 and 0.183 in normal, prediabetes and type 2 diabetes, respectively; the adjusted type 2 diabetes–normal coefficient was 0.191 on the arcsine-square-root scale (BH-FDR across eight clusters, 1.78×10^{-4}). C0 decreased from 0.242 to 0.137 (coefficient, -0.122 ; FDR, 0.0213). These proportions describe recovered beta cells, not absolute islet abundance. Beta-cell quality control, qualifying-donor support and donor-level composition are shown in Extended Data Fig. 5.

Counts were aggregated at donor level before expression modelling.^{8–9,11} We fitted edgeR quasi-likelihood models to 100,434 eligible gene–cluster entries.⁴⁵ The type 2 diabetes–normal contrast yielded 418 associations involving 225 unique genes (Fig. 5c). Reproduction of the source-atlas whole-beta workflow⁴⁴ yielded 511 genes; 136/225 state-resolved genes overlapped this set and accounted for 310/418 associations, all direction-concordant. Limma-voom sensitivity analysis^{46–47} yielded 444 associations, 311 shared with edgeR and no shared sign conflict. Reactome⁴⁸ and Hallmark⁴⁹ collections were tested competitively with CAMERA.⁵⁰ CAMERA showed significant negative competitive gene-set signals for fatty-acid-metabolism-associated gene sets in all eight clusters and oxidative-phosphorylation-associated gene sets in seven clusters, significant positive competitive gene-set signals for TNF–NF- κ B-associated transcriptional programs in six clusters, and significant negative competitive gene-set signals for mTORC1-signalling-associated gene sets in six clusters (Fig. 5d). Whole-beta sensitivity analyses and the invalid cell-level counterfactual are shown in Extended Data Fig. 4; differential-expression, competitive gene-set-enrichment and atlas-overlap diagnostics are shown in Extended Data Fig. 6.

Discussion

scDesignGuard formalizes scientific authorization before execution. The framework makes source identity, biological replication, target support and estimability explicit prerequisites and records where authorization stops. The released evaluator checks a completed contract, while upstream semantic construction and downstream donor-aware statistics provide the surrounding workflow.

The 15-family overlap-excluded benchmark demonstrated that the evaluator distinguishes valid, hard-invalid and structurally unresolved designs: valid designs proceeded, demonstrated invalidities were blocked and unresolved records remained non-evaluable. Exact state concordance was 15/15, with Wilson intervals reporting the precision of each frozen stratum-specific endpoint. The source-bound human consensus, checklist and evaluator use a shared project taxonomy; the benchmark establishes deterministic state concordance within that reference framework.

The four-case end-to-end evaluation showed that PROCEED reached quantitatively supported target aggregation and estimable models in two cases, while two prespecified early stops retained explicit source-binding or donor-replication reasons. This operational evaluation links contract decisions to the next computational stage under the frozen contrasts, thresholds and gate order.

Controlled challenge perturbations probe rule sensitivity, repository audits test handling of incomplete evidence, software checks establish deterministic contract evaluation and the beta-cell application demonstrates donor-aware use of the broader framework. The beta-cell results localize signals within GSE221156 and provide defined targets for independent replication and causal or clinical evaluation.

Semantic contract construction is the next validation target because complex source records, ontology relations and estimands require trained interpretation. The present benchmark evaluates state concordance after source-bound fields have been reviewed. A future external study can quantify de novo contract agreement, processing time and usability among independent analysts.

Methods

Framework and scientific states

A task contract binds immutable source identifiers and versions, the independent biological unit, condition and pairing fields, source-defined target labels and ontology relations, support thresholds, estimand definitions and evidence-object references. Trained review or separate upstream code constructs these fields. The released evaluator accepts the completed contract, validates its schema and exact matrix-contrast estimability, and maps supplied evidence booleans to PROCEED, ABSTAIN, BLOCK or NON_EVALUABLE with ordered reasons. ABSTAIN was retained as a prespecified framework state for modelled uncertainty; its performance is a target for future empirical evaluation because no final benchmark or end-to-end case activated it.

Evidence tiers and decision hierarchy

Tier A requires at least five donors per group or five complete pairs, at least 20 target cells per donor, exact donor mapping, EXACT or CHILD target ontology, established analysis access and a complete estimand. Tier B requires at least three donors per group or three complete pairs and at least ten target cells per donor and is limited to supportive robustness or direction. Tier C covers metadata, provenance, repair and terminal-state evaluation. Tier D comprises known-invalid transformations of public real-data contracts. These project-defined cut-offs and the resulting support fields are computed upstream and supplied to the evaluator.

Terminal state was selected using the frozen precedence order BLOCK, NON_EVALUABLE, ABSTAIN and PROCEED. All activated reason codes were ordered first by terminal-state precedence and then lexicographically by code; the first code was reported as the primary reason and the remaining codes as ordered secondary reasons. In the broader framework ledger, unperformed or stage-inapplicable checks were reported as NOT_ASSESSED or NOT_APPLICABLE; these ledger labels are not accepted as boolean inputs by the current evaluator schema. Supplementary Table 3 documents the 21-code NM03 v0.1.0 evaluator registry; frozen benchmark-compiler reason labels used for the 15-family benchmark are reported separately in Supplementary Table 1.

Final benchmark selection and human consensus

Two human reviewers verified prepared source-bound records while blinded to evaluator, executable-checklist, end-to-end and biological-outcome outputs; disagreements were resolved and the consensus labels were frozen. The 24 frozen candidates were subsequently audited against a project-wide family-identity registry linking repository accessions or collection identifiers with publication, project and cohort aliases. Nine families with identified development or calibration study-family overlap were excluded without replacement before final endpoint estimation, leaving 15 families. Selection was based on family identity and project role before evaluator, accessibility and biological-result review. The resulting reference is termed frozen source-bound human consensus. De novo contract-construction agreement was outside the prespecified endpoints.

Benchmark endpoints

VALID_DESIGN, HARD_INVALID_DESIGN and STRUCTURALLY_UNRESOLVED reference classes implied expected PROCEED, BLOCK and NON_EVALUABLE states, respectively. Primary endpoints were valid clearance, hard-invalid detection, unresolved containment, unsafe continuation and exact state concordance. Each rate is reported with exact membership and a two-sided 95% Wilson interval. Evaluator and executable-checklist outputs did not modify the human reference.

End-to-end target-support and model-validity evaluation

A separate four-case overlap-excluded cohort was processed in frozen order through source scope, structural design, target support, count aggregation, model validity and final authorization. Candidate inclusion did not use scDesignGuard state, downstream effect estimates or biological outcomes.

Unpaired comparisons required at least five qualifying donors per condition; paired comparisons required at least five complete pairs. Each donor-target unit required at least 20 target cells and a finite positive raw integer library. Repeated samples were aggregated to donors.^{8–9,11} Model validity required a full-column-rank matrix, an estimable condition effect and positive residual degrees of freedom. A Python implementation computed donor assignment, target-cell counts, raw-library sums, pseudobulk dimensions, rank, estimability and residual degrees of freedom. Count quantities were recomputed by an independently compiled C implementation and model-matrix quantities by an independent R implementation. The gate evaluation concluded at final authorization.

Controlled challenge perturbations and repository transport

Nine transformations modified one prespecified property while retaining all other contract fields: donor deletion, cell-level pseudoreplication, batch-condition confounding, duplicate donor, pair destruction, wrong target, wrong version, mixed strata and low support. Expected states and primary reasons were fixed before evaluation.

Repository transport used official metadata and identity endpoints for CELLxGENE,³⁷ NCBI GEO³⁸ and BioStudies/ArrayExpress.³⁹ Single Cell Portal and Human Cell Atlas sources were queried through their official metadata interfaces; the Human Cell Atlas context is described elsewhere.⁴⁰ Authentication failures, absent row metadata and unresolved donor mappings were retained as evidence states. Expression and outcomes were outside the transport-audit protocol.

Software reproducibility

The evaluated NM03 v0.1.0 software was frozen as a Python wheel, OCI image, JSON Schema, 21-code reason registry and software bill of materials. Public packaging release v0.1.1 retains these objects unchanged and adds archive-relative tests, four synthetic terminal-state fixtures with golden outputs, 17 source test methods, four clean-environment wheel fixture checks and checksum-bound integrity receipts. The release is available at <https://github.com/wang17777/scDesignGuard/releases/tag/v0.1.1>.

Pancreatic beta-cell analysis

The exact CELLxGENE object³⁷ for GSE221156⁴⁴ contained 99,029 author-annotated beta cells from 48 donors. Author C0–C7 labels were retained without marker-based renaming. Raw counts were summed once per donor–cluster. A donor–cluster unit required at least 20 cells and a positive raw library sum.

Composition models used arcsine-square-root donor proportions and adjusted for assay, sex, ethnicity, age and body mass index; inference used limma empirical-Bayes tests⁴⁷ with BH correction across eight clusters per contrast. Counts were aggregated at donor level to preserve the biological unit.^{8–9,11} Cluster-specific models were fitted with edgeR quasi-likelihood methods⁴⁵ using the same covariate design. Genes were filtered within cluster, and BH correction was applied across all eligible gene–cluster tests within each contrast. Limma-voom^{46–47} used identical matrices, covariates and contrasts. All eligible Reactome⁴⁸ and Hallmark⁴⁹ gene sets were tested competitively with CAMERA,⁵⁰ with BH correction across the complete cluster–gene-set universe within each collection and contrast. For Fig. 5d, the 12 Hallmark gene sets with the smallest minimum global cluster–gene-set FDR across C0–C7 in the type 2 diabetes–normal contrast were displayed. Extended Data Fig. 6 applied the same ranking rule separately to display 15 Hallmark and 20 Reactome gene sets. Display selection did not change either multiple-testing universe.

The atlas-overlap analysis reproduced the GSE221156 authors' whole-beta type 2 diabetes–normal workflow,⁴⁴ yielding 511 genes. State-resolved genes and gene–cluster associations were joined to that set in a shared 13,207-gene test universe. Overlap denominators were 225 unique genes and 418 significant associations; direction concordance was evaluated only for overlapping associations.

Statistics and reporting

Cells were never treated as independent biological replicates.^{8–10} Binomial intervals are two-sided Wilson intervals. Differential-expression and composition P values are two-sided and adjusted as specified above. Competitive enrichment direction is encoded by the sign of the CAMERA test statistic. No threshold, endpoint or truth rule for the final 15-family overlap-excluded benchmark or the four-case end-to-end evaluation was modified after those analysis sets and endpoint definitions were frozen.

Use of large language models

OpenAI Codex supported language editing, cross-file consistency checks and software-archive quality control during manuscript preparation. The authors reviewed and verified every retained suggestion and take responsibility for the final content. Experimental data, statistical results and scientific conclusions were generated and validated by the authors.

Data availability

The final 15-family benchmark membership, four-case end-to-end gate results, controlled-challenge results, repository-transport summaries, quantitative reconciliation and figure Source Data accompany the manuscript. Historical development and audit objects are retained under version control but are not part of the submitted scientific denominators. Expression objects are not redistributed and remain subject to repository and study terms. The same-cohort biological application is available under accession GSE221156.

Code availability

The evaluated NM03 v0.1.0 source code, wheel and OCI asset are publicly available in the v0.1.0 release. The checksum-bound v0.1.1 reproducibility release is available at <https://github.com/wang177777/scDesignGuard/releases/tag/v0.1.1> and provides archive-relative tests, four terminal-state fixtures, golden outputs and integrity receipts. Evaluator source, JSON Schema, 21-code reason registry, wheel and OCI identities are unchanged. The software is distributed under the BSD 3-Clause license.

Ethics statement

This study exclusively performed secondary analyses of publicly available, de-identified human datasets. No new participants were recruited, no intervention was performed, no new biological specimens were collected, and the investigators had no access to identifiable private information. Accordingly, under applicable institutional policies, additional ethics committee approval and informed consent were not required for this secondary analysis. Ethics approvals and informed-consent or waiver procedures for the original data collections are reported in the respective source publications and repositories. The secondary use and reporting of the data followed applicable repository terms, institutional policies and the relevant ethical principles of the 2024 Declaration of Helsinki.

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Author contributions

Guoyong Wang, Kaijun Zhang, Jiyue Jiang, Yu Li and Xiaonan Yang conceived the study. Kaijun Zhang, Jiyue Jiang and Yu Li developed the methodology. Jiyue Jiang and Yu Li developed the software, curated the data and performed the formal analysis. Guoyong Wang, Weixin Wang, Chaonan Wang, Hui Bi, Haojun Liang, Zuoliang Qi and Ying Huang contributed to clinical and biological interpretation. Guoyong Wang, Kaijun Zhang, Jiyue Jiang and Yu Li drafted the manuscript. All authors reviewed, edited and approved the final manuscript. Guoyong Wang and Xiaonan Yang

supervised the study. Xiaonan Yang and Chaonan Wang acquired funding. Guoyong Wang, Kaijun Zhang, Jiyue Jiang and Xiaonan Yang contributed equally to this work.

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Competing interests

The authors declare no competing interests.

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Figure legends

Figure 1 | Source-bound design evidence and stage-specific authorization. a, Upstream review compiles source identity, donor and condition structure, target ontology, support and estimand into an evidence-bearing contract; the released evaluator begins from this completed contract and returns PROCEED, ABSTAIN, BLOCK or NON_EVALUABLE with explicit reasons. b, The broader framework comprises source scope, structural design, target support, count aggregation, model validity and inference authorization; each passed gate authorizes assessment of the next. c, GSE144735 worked example: six complete pairs, 3,356 target epithelial cells, 12 donor-condition pseudobulks, design-matrix rank 7/7 and five residual degrees of freedom. The panel reports the evidence supporting final authorization.

Figure 2 | Overlap-excluded benchmark and state concordance. a, Frozen source-bound human consensus was established before a project-wide family-identity audit defined the final overlap-excluded endpoint set. b, Frozen human-reference strata: four valid, six hard-invalid and five structurally unresolved families. c, Exact proportions and two-sided 95% Wilson intervals for valid clearance (4/4), hard-invalid detection (6/6), unresolved containment (5/5), unsafe continuation (0/6) and exact state concordance (15/15). d, Family-level state matrix. These endpoints measure concordance within the frozen source-bound human reference.

Figure 3 | Overlap-excluded end-to-end outcomes. a, Outcomes for four cases. GSE144735 and GSE123813 passed all gates; Human Cell Atlas project 135f7f5c-4a85-4bcf-9f7c-4f035ff1e428 stopped at structural source binding and GSE81076 stopped at the donor-replication threshold. b, Exact quantitative checks for the two full gate passes. GSE144735 comprised six complete pairs, 3,356 target cells, 12 pseudobulks, rank 7/7 and five residual degrees of freedom; GSE123813 comprised 11 complete pairs, 33,106 target cells, 22 pseudobulks, rank 12/12 and ten residual degrees of freedom. c, Number of cases clearing each sequential gate. The evaluation endpoint was final authorization.

Figure 4 | Controlled challenge perturbations, repository audits and software reproduction. a, Nine deterministic challenge perturbations; five returned BLOCK and four NON_EVALUABLE, and all nine matched the prespecified primary reason. b, Eighteen Tier C audits across BioStudies/ArrayExpress (n=5), Single Cell Portal (n=4), NCBI GEO (n=5) and Human Cell Atlas (n=4). All received NON_EVALUABLE outcomes for unresolved ontology, donor-mapping and support evidence. c, Public packaging release v0.1.1 completed 17/17 source test methods and four terminal-state

fixture subtests with zero failures, errors or skips; the unchanged v0.1.0 wheel passed 4/4 fixture CLI checks in a clean temporary environment. These checks cover evaluator software and supplied synthetic fixtures.

Figure 5 | Donor-aware beta-cell composition, transcription and competitive gene-set enrichment. a, Mean donor proportions of author-defined clusters C0–C7 in normal (ND; n=17 donors), prediabetes (PD; n=14) and type 2 diabetes (T2D; n=17); bars are arithmetic means across donors. b, Adjusted T2D–ND composition coefficients from donor-level models on the arcsine-square-root scale after adjustment for assay, sex, ethnicity, age and body mass index. Points are coefficients; error bars are descriptive ± 1.96 unmoderated coefficient standard errors (residual d.f.=39). P values are two-sided limma empirical-Bayes tests; FDR values are Benjamini–Hochberg adjusted across eight clusters. Positive values indicate higher recovered-cell proportion in T2D. c, Significant gene–cluster associations after BH correction across 100,434 eligible tests in the T2D–ND contrast. Bars show edgeR quasi-likelihood and limma-voom counts. edgeR identified 418 associations involving 225 genes; 136/225 genes overlapped the same-cohort 511-gene whole-beta atlas set, accounting for 310/418 associations with 310/310 direction agreement. d, Signed $-\log_{10}$ global FDR for competitive Hallmark CAMERA gene-set tests. All eligible Hallmark cluster–gene-set tests were included in BH correction; for visualization, the 12 gene sets with the smallest minimum global FDR across C0–C7 were displayed. Red denotes positive and blue negative competitive gene-set signals in T2D. Display selection did not alter the inferential universe.

Extended Data figure legends

Extended Data Figure 1 | Development support, robustness and calibration. a, Qualifying independent-unit support under the development protocol; the dashed line marks the project-defined five-donor cut-off. b, Leave-one-donor-out direction-stability profiles. c, Calibration-label composition. d, Minimum target-cell support among qualifying donor-target units; the dashed line marks the project-defined 20-cell cut-off. These panels summarize development evidence.

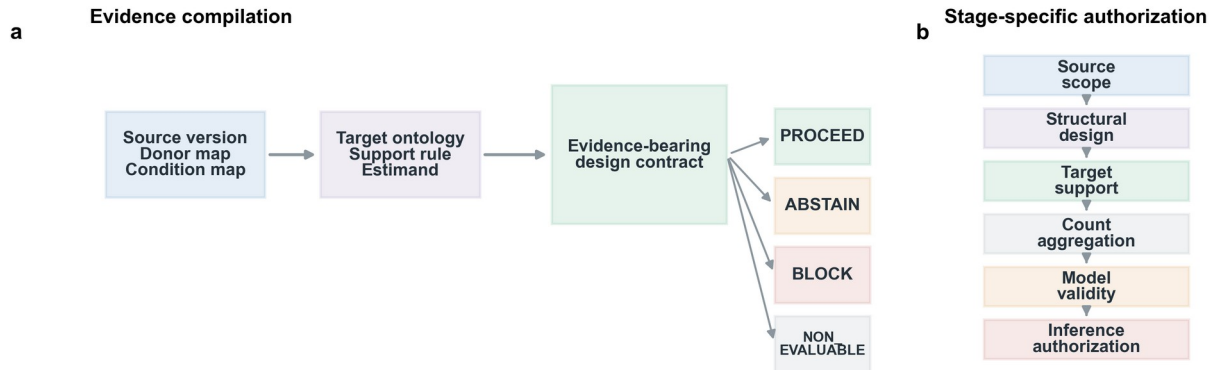
Extended Data Figure 2 | Synthetic uncertainty and software archive status. a, Descriptive Monte Carlo estimates for pooled Type-I rate and mean replicate FDP/FDR with simulation intervals and Monte Carlo standard errors. The 0.05 line is the prespecified reference. b, Public release v0.1.1 adds four fixtures, golden outputs and archive-relative tests while retaining the exact v0.1.0 wheel and OCI hashes. Source tests and clean-environment wheel fixture checks passed, and the SBOM receipt verifies file integrity. The release is available at <https://github.com/wang17777/scDesignGuard/releases/tag/v0.1.1>.

Extended Data Figure 3 | Controlled challenge perturbations and repository transport. a, Nine controlled challenge perturbations; five returned BLOCK and four NON_EVALUABLE, with observed terminal states and primary reasons. b, Tier C transport-audit families by repository and time stratum. c, Matched clean–challenge contract checks. Transport audits used identity and metadata endpoints; their endpoint was terminal-state evaluation.

Extended Data Figure 4 | Whole-beta robustness and invalid counterfactual. a, Leave-one-donor-out effect correlations. b, Direction-retention fractions. c, Filtering sensitivity. d, Cell-level pseudoreplication counterfactual illustrating the design violation and its statistical consequences.

Extended Data Figure 5 | Beta-cell quality control, support and composition. a, EdgeR multidimensional-scaling coordinates for donor–cluster pseudobulks, coloured by condition; no point was excluded after inspection. b, Qualifying donor counts under the ≥ 20 -cell rule. c, Donor-level recovered-beta-cell composition. Box plots show medians, interquartile ranges and whiskers extending to $1.5\times$ the interquartile range.

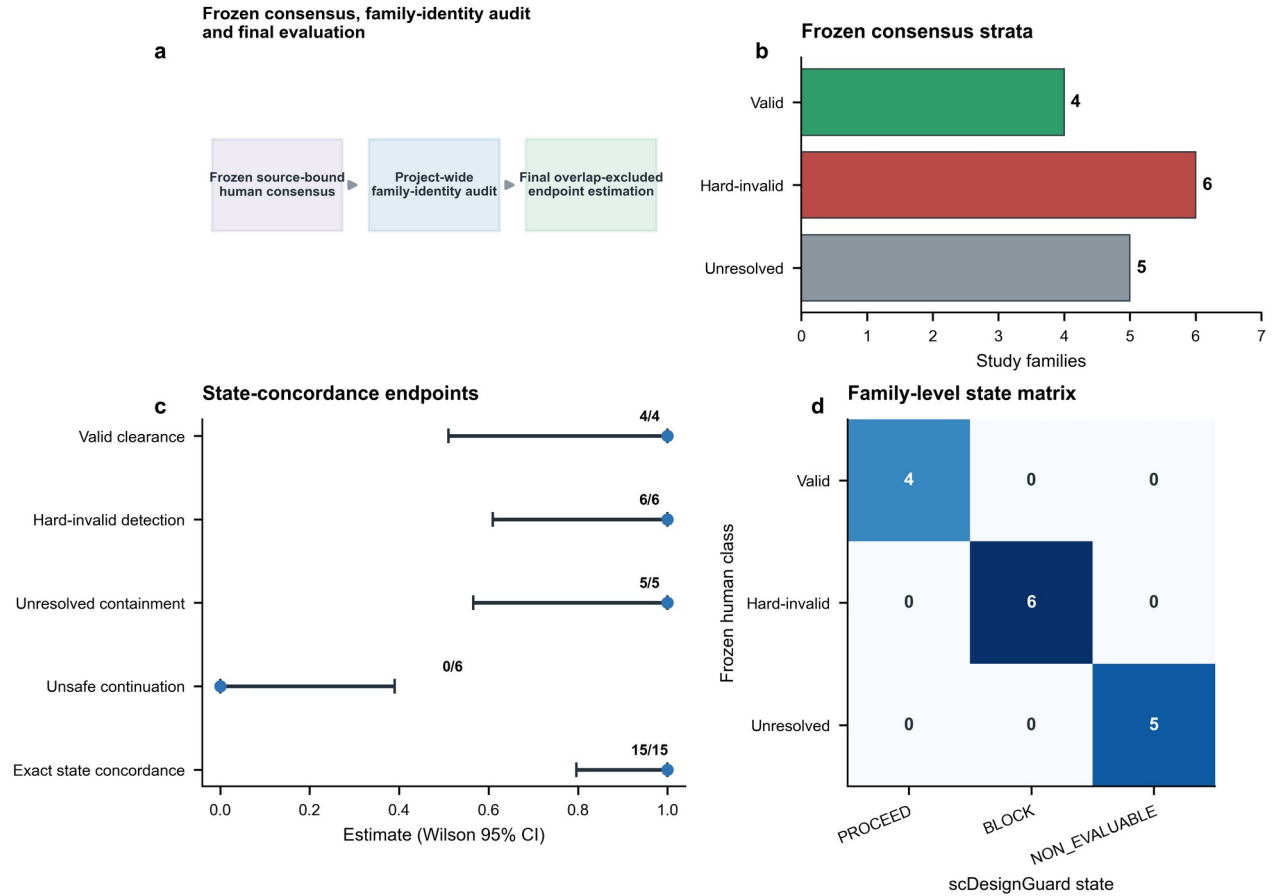
Extended Data Figure 6 | Beta-cell differential expression, competitive gene-set enrichment and atlas overlap. a, EdgeR quasi-likelihood T2D–ND effects; red points pass BH correction across all eligible gene–cluster tests. b, c, Signed $-\log_{10}$ global FDR for Hallmark and Reactome CAMERA gene-set tests. The 15 Hallmark and 20 Reactome gene sets with the smallest minimum global cluster–gene-set FDR across C0–C7 were displayed separately; all eligible tests in each collection remained in the BH correction universe. d, edgeR–voom log-fold-change concordance. e, Same-cohort overlap: 136/225 state-resolved genes and 310/418 associations overlap the whole-beta atlas set, with 310/310 direction concordance. Genes outside the atlas overlap are candidates for independent validation.



c Worked example: GSE144735

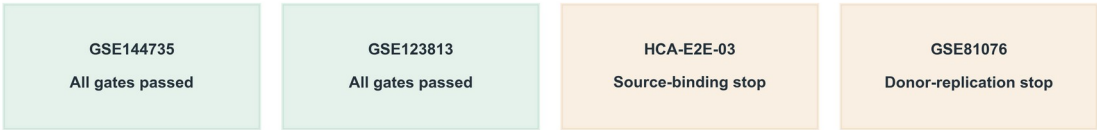


Frozen consensus, family-identity audit and final evaluation



a

Final gate outcome



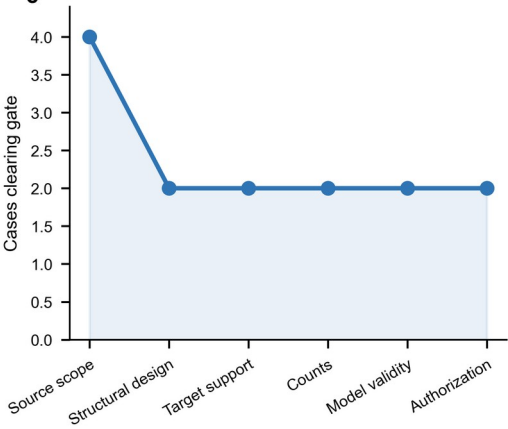
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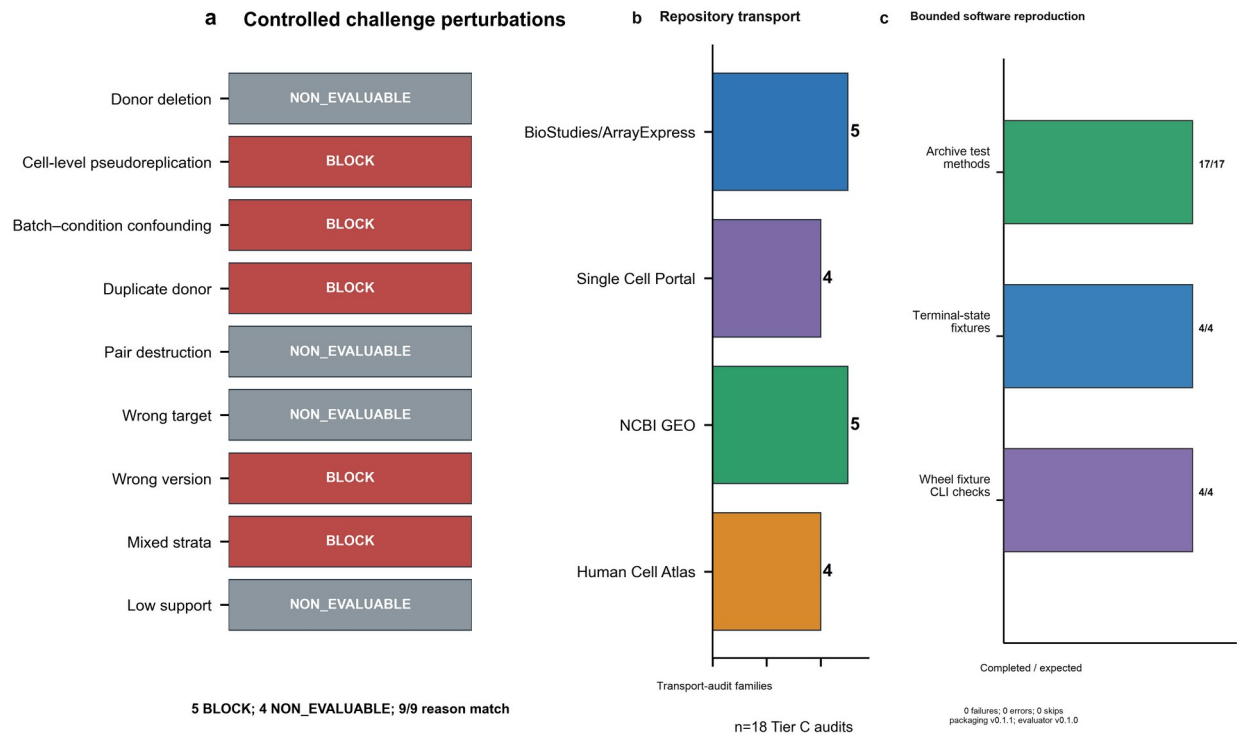
Quantitative gate checks

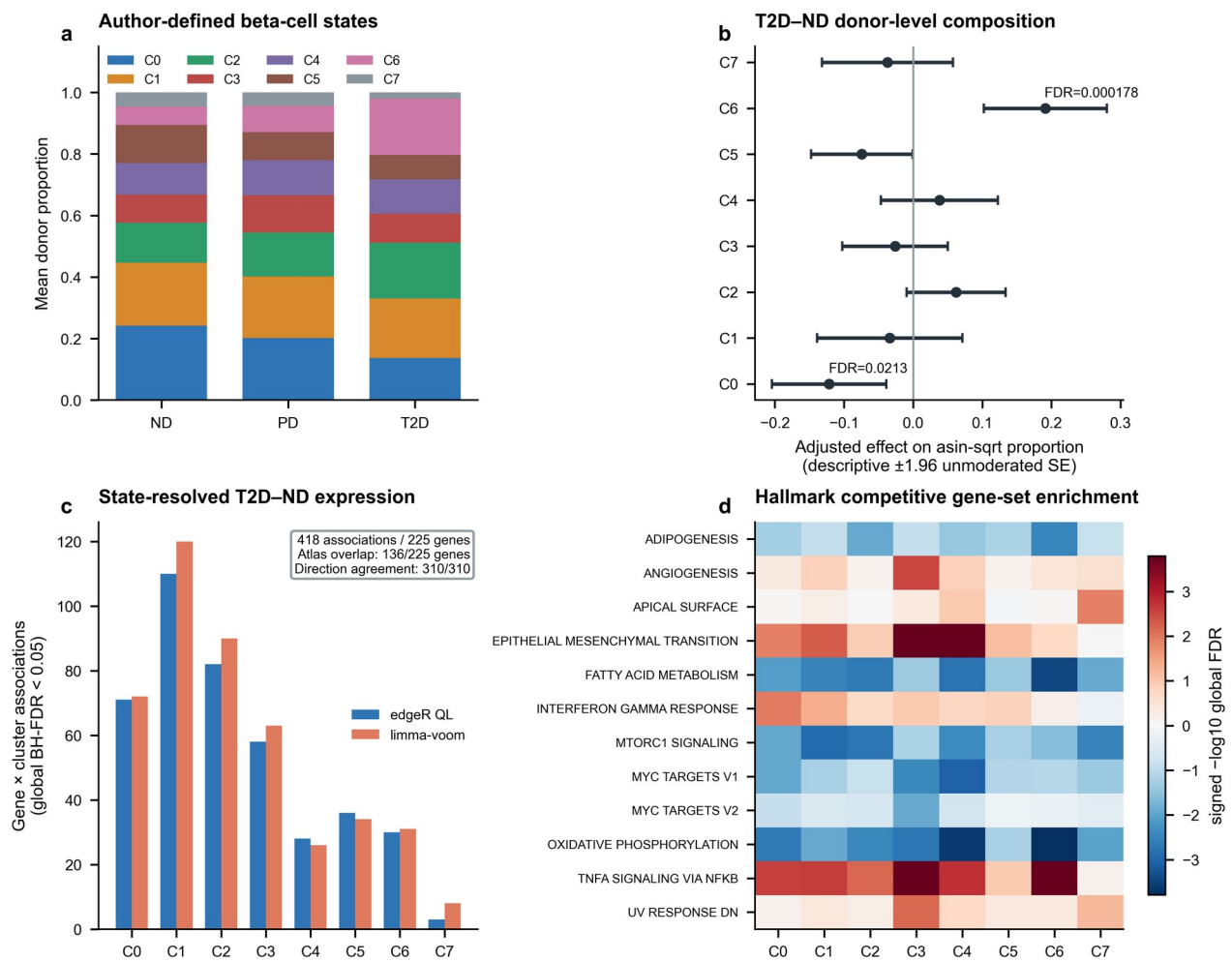
	Complete pairs	Target cells	Pseudobulk rows	Design rank	Residual d.f.
GSE144735	6	3,356	12	7/7	5
GSE123813	11	33,106	22	12/12	10

c

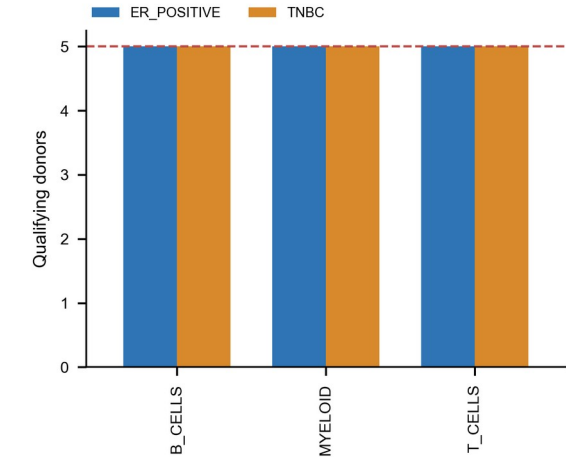
Cases clearing each sequential gate



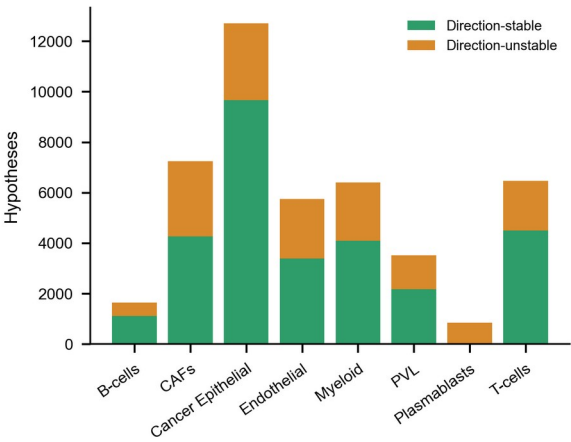




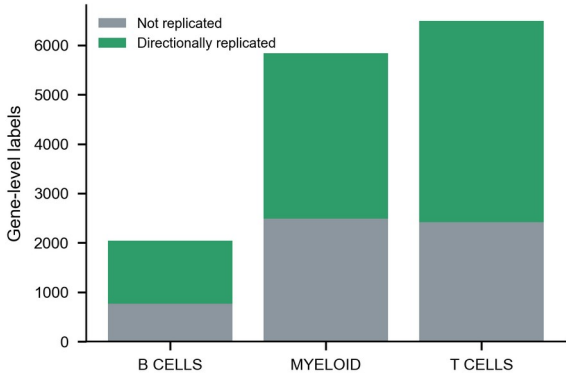
a Independent-unit support



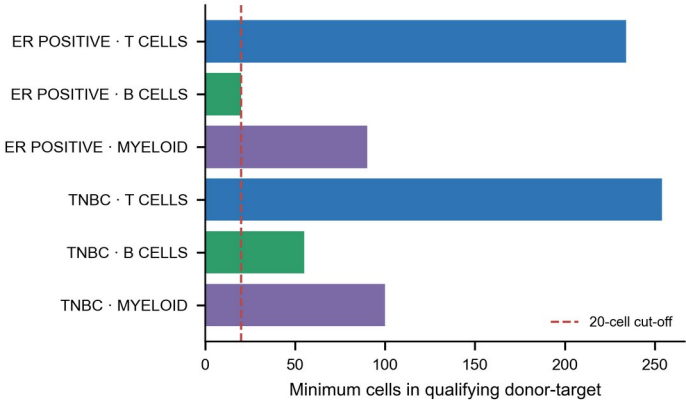
b Leave-one-donor-out diagnostics

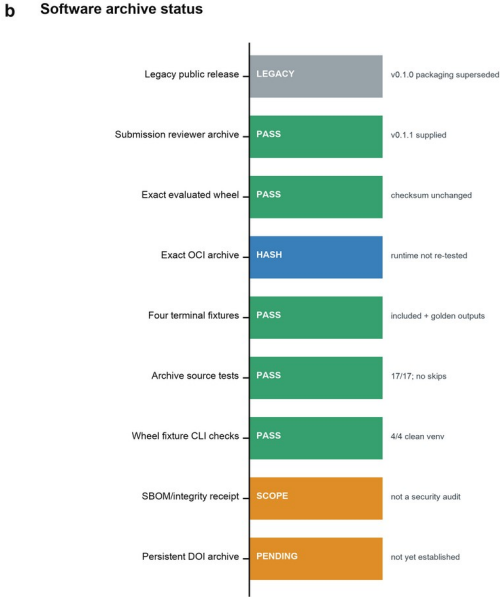
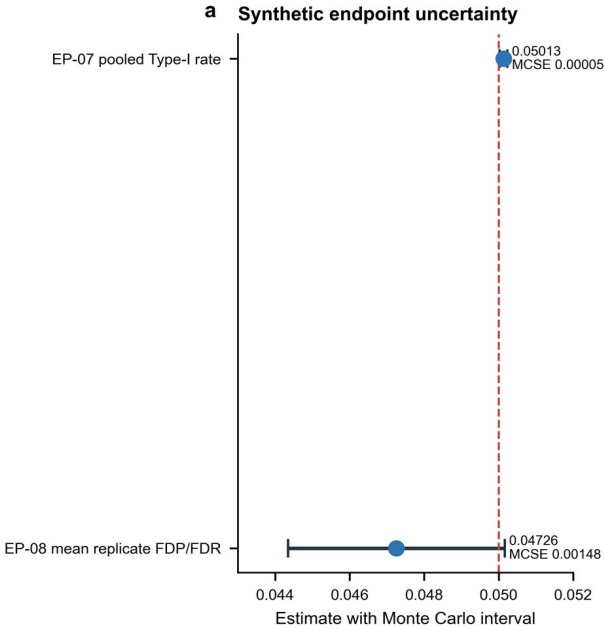


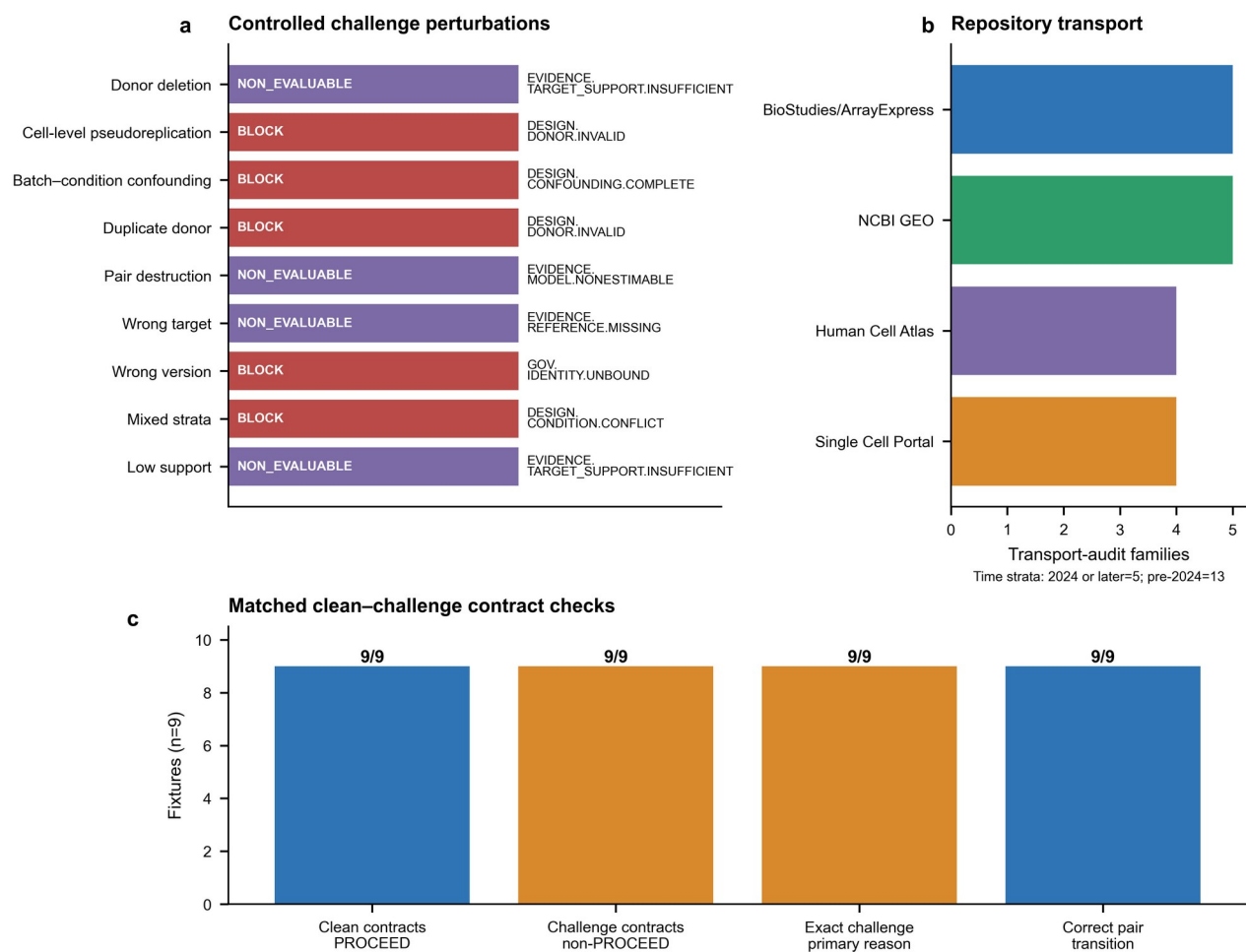
c Calibration-label composition

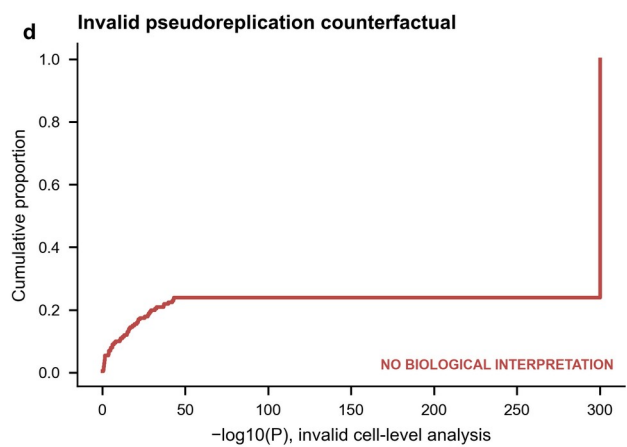
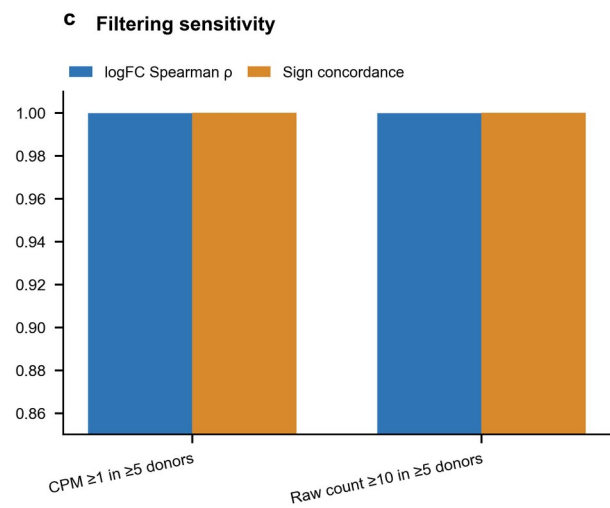
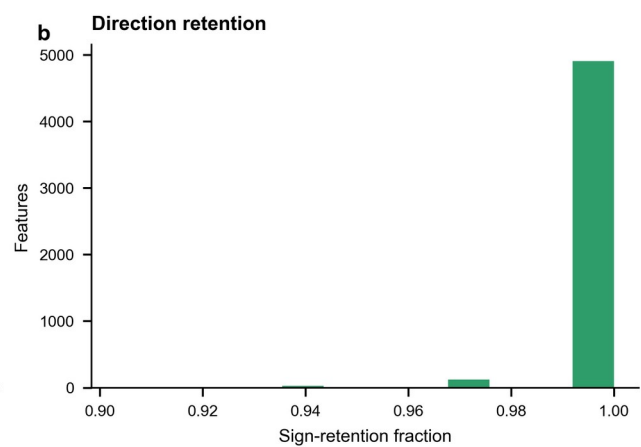
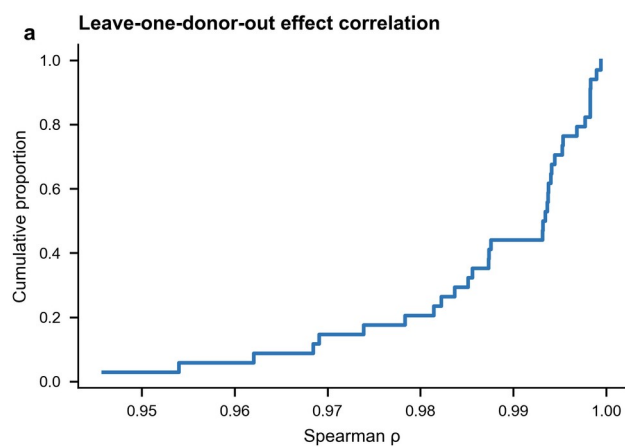


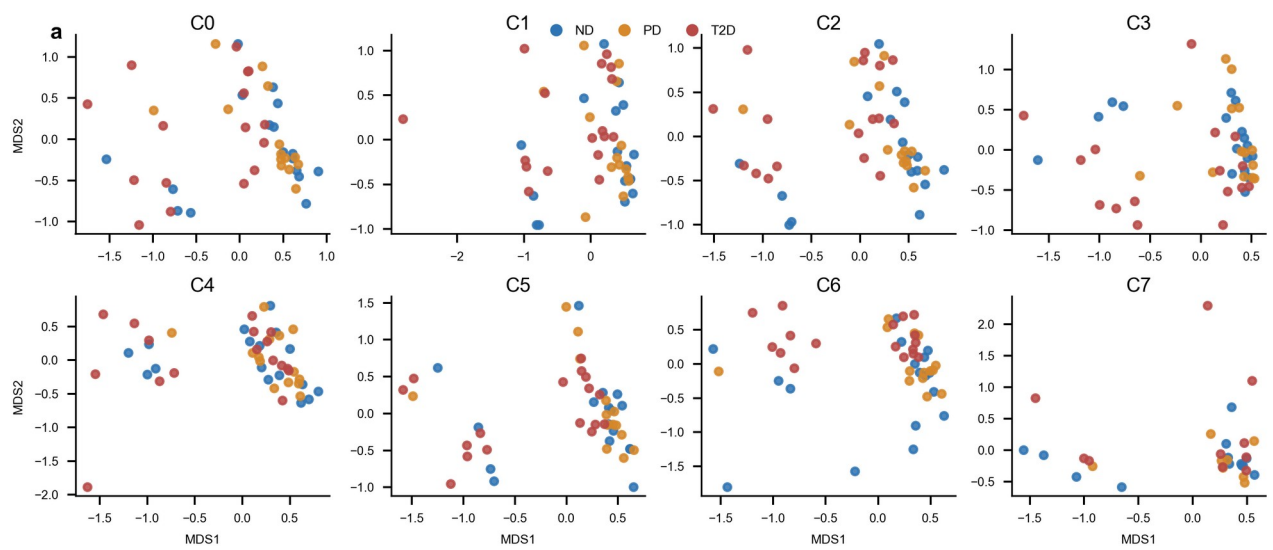
d Target-support floor











b Qualifying donors (≥ 20 cells)

	ND	PD	T2D
C0	17	14	16
C1	17	14	17
C2	17	14	17
C3	17	14	16
C4	17	14	17
C5	17	14	17
C6	17	14	17
C7	13	8	10

c Recovered beta-cell composition

